



Wasim Ansari

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🚩 Indian

📍 Dehradun, India

🌐 LinkedIn Profile

🐙 GitHub

🏆 Kaggle

🔗 Portfolio

📄 Publications

Summary

I help organisations build scalable GenAI and ML systems that automate operational workflows and take smarter decisions. With IIT Madras training and eight years of cross-domain experience, I deliver production ready Machine Learning models, RAG chatbots, multi-agent systems, and other AI services. My expertise lies in delivering measurable business value with complete observability that are ethically aligned and compliant.

Projects

AI Travel Planner (Multi-Agent Orchestration using CrewAI with online search tool) [🔗](#)

- **Summary:** Built a multi-agent AI orchestration where three coordinated agents autonomously research tourist destinations, analyze user preferences, and generate personalized end-to-end travel plans.
- **Technologies:** CrewAI, LangChain, Streamlit, Python, CI/CD.

MediFlow - Medical AI Assistant (Multi-Agent Chatbot with MCP RAG tool and custom tools) [🔗](#)

- **Summary:** A medical assistant chatbot that delivers personalized post-discharge care through multi-agent workflow. Receptionist with patient history tool & Clinical Agent with Nephrology RAG and search tools.
- **Technologies:** LangChain, LangGraph, LangSmith, MCP client, Google Cloud Run, CI/CD, HTML, React js.

Nephrology RAG MCP Server - exposing Nephrology RAG services as streamable HTTP [🔗](#)

- **Summary:** Designed an MCP server exposing domain-specific retrieval-augmented generation (RAG) services as streamable HTTP endpoints for seamless integration with multiple AI agents concurrently.
- **Technologies:** FAISS, MCP Server, Python, Asyncio, Google Cloud Run, Docker, JSON-RPC 2.0

DocuDroid - AI Document Analysis Bot (LLM + Vector Search for Smart Document QA) [🔗](#)

- **Summary:** Implemented a temporary in-memory vector store for immediate retrieval, combined with LLM-generated answers. Useful to chat instantly with the PDFs or static documentation webpages.
- **Technologies:** Python, FastAPI, LangChain, InMemoryVectorStore, Docker, Google Cloud Run.

PDF Excel Extractor - AI-Power-Tool (extracts data from PDFs and provided structured Excel) [🔗](#)

- **Summary:** LLM analyzes documents to determine relevant fields for extraction (no hardcoded schema). Converts extracted text into structured Excel spreadsheets for downstream task. Complete Automation.
- **Technologies:** PDF Scraping, LLM schema detection, FastAPI, BLEU Score, Docker, Google Cloud Run.

Flux - Text-to-Image Generator (Powered by Hugging Face Diffusion Model) [🔗](#)

- **Summary:** A powerful image generation tool leveraging state-of-the-art HuggingFace diffusion models to create images from text prompt. Offers intuitive UI as well as API to connect with other services.
- **Technologies:** Python, azurewebsites, Hugging Face Diffusers, PIL, Docker, FastAPI.

Jurisol - Legal AI Assistant (LLM based RAG application with online search and summarization tools) [🔗](#)

- **Summary:** Engineered an intelligent legal search assistant using Retrieval-Augmented Generation (RAG), Chromadb vector database. Equipped with summarization tool and online legal document search.
- **Technologies:** RAG, Vector Database, LangGraph, FastAPI, Mistral LLM, Document Summarization.

BikeValuePro - ML Model for price prediction (Machine Learning Model with 94.9% Accuracy) [🔗](#)

- **Summary:** Trained a ML model for used bike prices using Scikit-learn and XGBoost. Includes smart Streamlit UI and deployment-ready architecture. Display personalized advice based on predicted price.
- **Technologies:** Machine Learning, Predictive Analytics, XGBoost, Scikit-learn, Pandas, Streamlit.

IITM Assignment Solver (Problem solving using Code Generation and Self Executing AI Agent)

- **Summary:** Built an AI agent to process ZIP, CSV, PDF, Excel, and JSON files, generate Python code, and run it in a sandboxed environment. Developed to participate in an IITM college challenge securing A+ Grade.
- **Technologies:** AI Agent, Code Generation, LLM, File Processing, FastAPI, LangChain, Mistral, Automation.

Top Skills

Data Analysis & Exploration

- Statistical Analysis (Python, Pandas, NumPy, Metabase, OpenRefine, Hypothesis Testing)
- Visualization (Matplotlib, Seaborn, PowerBI)

Generative AI & Agentic AI

- Frameworks (LangChain, LangGraph, CrewAI, Agno, MCP clients and servers)
- Vector Databases (ChromaDB, FAISS, Neo4j)
- Workflow Orchestration & Automation (n8n)

Cloud & DevOps Platforms

- Google Cloud (Cloud Run, Cloud SQL, Cloud Storage, Cloud Build, SDK)
- Azure (Container Apps, App Service / Web Apps)
- Vercel, GitHub Pages, Hugging Face Spaces

Machine Learning & Deep Learning

- ML Algorithms (Supervised & Unsupervised)
- Neural Networks (Computer Vision, Finetuning Transformers, Custom Modelling)
- Frameworks (Scikit-Learn, Keras, PyTorch, PyTorch Lightning, HuggingFace Transformer)

MLOps & Model Deployment

- Model Serving (Docker, FastAPI, Celery, Redis)
- CI/CD & Versioning (Git, GitHub Actions)

Monitoring & Observability

- Model Performance Tracking (Grafana, WandB, TrackIO, LangSmith)
- Drift Detection & Automated Retraining
- Guardrail, Prompt Experimentation

Experience

12/2025 – Present
Hyderabad, India

TechPranee (SpearHub)

AI/ML Engineering Intern

- Designed and deployed production-ready AI solutions, **integrating advanced ML and Gen AI capabilities** into existing enterprise systems to enhance product functionality and business outcomes.
- **Collaborated cross-functionally with engineering teams** to understand problem statements, plan execution, and ensure timely, high-quality delivery.
- Led rapid **experimentation cycles, performing iterative analysis**, and presenting actionable insights and recommendations to project leaders.
- Built and successfully deployed: Time-series forecasting model integrated into ERP systems, **Hybrid RAG chatbot (Neo4j Graph + Vector Search)** for HR automation, Ongoing: Multi-agent system for intelligent document review
- Ensured system reliability by **optimizing performance for high load and concurrent usage** in production environments.
- Owned **debugging, testing, and continuous improvement** of deployed systems to maintain stability and scalability.

09/2024 – 12/2025
Dehradun, India

WSMAISYS (Independent Freelancer)

AI/ML Engineer & Gen AI Architect

- Pursued IIT Madras Diploma in Data Science, with a focus on **Deep Learning, Generative AI, and cloud-based machine learning** systems.
- Delivered freelance AI/ML solutions for multiple clients, including **predictive modeling, AI-powered tools, workflow automation** using n8n, and operational efficiency enhancements.
- Built a diverse portfolio of real-world projects demonstrating strong **problem-solving ability, technical versatility, and hands-on experience** across end-to-end AI/ML pipelines.

12/2022 – 09/2023
Kingston Upon Hull,
UK

Booze Blitz - Ahaan Limited

Retail Sales Manager

- Led compliance and strategic sales planning that boosted revenue by 20%.
- Inspired ML-based solutions for pricing, demand forecasting and **retail analytics**.

09/2021 – 11/2022 Dehradun, India	Ahlan Exports Import Export Entrepreneur - Managed international B2B logistics, compliance and digital marketing funnels. - Experience translates into AI interest in supply chain optimization.
02/2015 – 01/2020 Dubai, UAE	FAHZAM Properties Real Estate Consultant - Managed property portfolios & client legal clearances. Certified RERA Broker. - Insights now inform AI projects in real estate valuation and docs automation.
09/2013 – 04/2014 Hyderabad, India	QuisLex Legal Services Legal Associate - Reviewed legal documentation & case summaries with accuracy and compliance. - Inspired the need for automation in document review and compliance QA via AI.
12/2012 – 07/2013 Pune, India	Mindcrest India Research Associate-Citator - Classified legal cases with 90%+ accuracy in half the average processing time. - Domain insights now power AI-based legal summarizers and DB management.

Education

09/2024	Indian Institute of Technology, Madras ↗ Diploma in Data Science Pursuing Coursework: ML, Linear Algebra, Statistics, Business Analytics, Tools for Data Science, Deep Learning, Gen AI, Capstone Projects. Grade: CGPA 8+ Projects: System Threat Forecaster (Top 5% in Kaggle), Assignment AI Agent, Protein Secondary Structure Prediction, Music Genre Classification.
2007 – 2012	Savitribai Phule Pune University ↗ Bachelor of Laws (5 years integrated) Coursework: Constitution, Civil and Criminal Laws, Procedural Laws, IPR, Banking Law, Environment Law, Drafting. Grade: Higher Secondary Class Achievements: Participant of 1st Student Parliament, Moot Court Competitions.

Languages

English — IELTS 7 Band	Hindi — Native or Bilingual	Urdu — Native or Bilingual
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Certifications

Understanding Google Cloud Platform ↗	Azure Essentials ↗	Dubai RERA Certification ↗
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Thoughts and Publications

- Adapting Export Strategies for Indian Handmade Carpets: A Response to Rising Global Tariffs** [↗](#)
Discover a data-driven export strategy for India's handmade carpet industry navigating rising U.S. tariffs with advanced Python-led trade analytics, market opportunity scoring, and competitive benchmarking.
- Hallucinations in AI Legal Research: A Lawyer's Worst Nightmare?** [↗](#)
Authored an article on AI hallucinations in legal research, highlighting real-world risks of fabricated case law and the need for prompt precision. Advocated safe AI adoption in law through domain-specific models and ethical safeguards.
- Teaching Computers to See Like Humans: The Brain Science Behind Smart Technology** [↗](#)
The article explains how Convolutional Neural Networks (CNNs) work - using kernels, feature maps and layering - to enable machines to "see" images like humans, powering modern computer-vision applications.